Rahul Kolla

Charlotte, NC — +1 (704) 345-8984 — rahulkolla17@gmail.com — LinkedIn — GitHub — Portfolio

Professional Summary

Agentic AI engineer who builds multi-agent LangGraph systems that automate SAP processes and power enterprise chatbots at AiFA Labs (Cerebro SASA platform), cutting Functional Design Specification generation time by 43% and chatbot response latency by 60%. Experienced in SAP quality assurance and backend development, and strong in Python, TypeScript, FastAPI, LangChain, and LangGraph. M.S. in Computer Science, 2026.

Technical Skills

AI & LLM Systems: Autonomous AI Agents, Deep Agents, Multi-Agent Orchestration, ReAct, Tool and Function Calling, RAG Pipelines, Prompt and Context Engineering, LangChain, LangGraph, LlamaIndex, Claude API, OpenAI API, Ollama, Hugging Face, Vector Databases, GPT-5
Programming: Python, TypeScript, JavaScript, Java, SQL, Bash
Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, Microservices, JWT Authentication, PostgreSQL, SQLite, Firebase (Cloud Functions, Firestore), Next.js
DevOps / Infrastructure: Docker, AWS, GitHub Actions, CI/CD, Linux, Git/GitHub
AI / ML & Data: NLP, Deep Learning, CNN, LSTM, Pandas, NumPy, Matplotlib, Tableau

Experience

Software Intern — *Charlotte, NC*
AiFA Labs – Cerebro SASA Platform — **Sep 2025 – Present**

- Architected Deep Agents workflows in LangGraph and LangChain, using planning, sub-agent delegation, and tool calling to automate SAP processes and reduce Functional Design Specification (FDS) generation time upto 43%.
- Built a LangGraph-based multi-agent orchestration layer for an enterprise chatbot, cutting response latency upto 60%, with RAG-based Confluence retrieval and function-calling integration that lets Jira ticket-creation agents work in parallel.
- Designed and built an LLM-powered release-notes pipeline (TypeScript, Firebase Cloud Functions, GPT-5) that turns merged GitHub PRs and Jira tickets into customer-facing blog posts, with Confluence approval before publishing to a Next.js portal, per-client access control, and automated monthly runs via Cloud Scheduler.
- Executed functional and UI regression testing on SASA, investigated client-reported issues, performed root-cause analysis, and validated backend integrations across SAP BTP, ABAP, and S/4HANA systems, reducing the defect rate upto 50% and driving feature and UX enhancements.

Backend Developer Intern — *Hyderabad, India*
AiFA Labs — **Jan 2024 – Jul 2024**

- Developed backend services using Python, Flask, and REST APIs to support real-time, AI-driven data operations and internal platform workflows.
- Automated ingestion, processing, and logging workflows, improving operational efficiency by 30%.

Associate Technical Lead – WIOS Club — *Amaravati, India*
VIT-AP University — **Jan 2023 – Aug 2023**

- Mentored students in full-stack (MERN) development and coordinated technical hackathons.

Projects

Helmsman – Autonomous DevOps AI Agent *Python, Claude Tool-Use API, GitHub Actions, Docker, Redis, Slack API*
Built a ReAct-style agent that monitors repository health, dependency risk (OSV.dev CVE scans), CI failures, and PR activity, and automates issue creation and Slack reporting. Redis caching and token budgeting reduce LLM cost.

FERRET – Autonomous AI Research Agent *Python, LangChain, OpenAI API*
Built an agent that performs multi-source web research, synthesizes findings, and generates structured reports, with live progress tracking, batch execution, and PDF, Markdown, and JSON export.

AI-Powered Smart Document Intelligence System *FastAPI, LangChain, Ollama, SQLite*
Built a local RAG platform for semantic PDF querying, contextual summarization, and document retrieval that runs fully offline.

Additional: *Security Pipeline for SIEM and IDS* – automated Wazuh and Suricata workflows with Ansible for real-time threat detection and monitoring. *Email Spam Detection* – CNN and LSTM classifier integrated with Flask APIs.

Education

University of North Carolina at Charlotte — *Charlotte, NC*
M.S. in Computer Science — **2024 – 2026**
CGPA: 4.00 / 4.00

Vellore Institute of Technology – AP — *Amaravati, India*
B.Tech in Computer Science — **2020 – 2024**
CGPA: 8.86 / 10.0

Certifications

Hugging Face Agents Course — Microsoft Azure Fundamentals (AZ-900) — NVIDIA Deep Learning Institute — Google Prompt to Prototype